

Building an HLA Typer Using Asian-Specific Pangenomes

First Author¹ and Last Author² 

BioHackathon series:
DBCLS BioHackathon 2026
Matsuyama, Japan, 2026
[asian-hla](#)

1 First Affiliation 2 ELIXIR Europe 

Submitted: 18 Sep 2026

License:
Authors retain copyright and
release the work under a Creative
Commons Attribution 4.0
International License ([CC-BY](#)).

Published by [BioHackrXiv.org](#)

Introduction

The HLA/MHC region on chromosome 6 is the most polymorphic part of the human genome and the single most important genomic determinant of transplant compatibility, drug hypersensitivity and autoimmune and infectious-disease risk. Existing HLA genotyping tools — HLA*LA (Dilthey et al., 2019), T1K (Song et al., 2023), SpecHLA (DeepOmics Lab, 2023) and others — were developed and tuned mostly against reference panels (GRCh38, CHM13, HPRC) that are enriched for European, African and admixed American ancestries. Middle Eastern and East/Southeast Asian populations are comparatively under-represented, even though HLA allele and haplotype frequencies are strongly population-specific. A recent deep long-read study of the non-classical class I genes in 531 Japanese individuals (Ito-Naito et al., 2026) names this under-representation explicitly as an open problem in the field.

At BioHackathon Japan 2026 our goal was to build an HLA genotyping method specifically informed by an Asian- and Arab-enriched pangenome — in spirit close to SpecHLA's approach of typing against a personalized/graph-derived reference rather than a single linear one, but built on our own pangenome graph rather than SpecHLA's. Over the course of the week we (1) assembled and extended an Asian- and Arab-enriched HLA haplotype panel to 754 haplotypes; (2) built a whole-MHC Minigraph-Cactus pangenome graph from it; (3) tested a first, direct way of using that graph for genotyping — align, call variants, phase, take a consensus, then annotate — against an established direct-read typer (T1K) and real experimental truth, to learn where a naive graph-based approach helps and where it does not; and (4) used what we learned to start building the actual typer: a PanGenie-based genotyper that calls graph bubbles directly from short reads without going through an explicit consensus step, evaluated so far on an initial cohort of East and South Asian 1000 Genomes donors. Parts of this are complete and validated; the typer itself is still in progress, and we report its status honestly alongside the completed pieces below.

Methods

An Asian- and Arab-enriched HLA haplotype panel (754 haplotypes)

We extended our original 610-haplotype panel (APR 106, HPRC r2 464, JaSaPaGe Saudi 18 + Japanese 20, GRCh38, CHM13) with 144 more haplotypes contributed by two additional pangenome projects whose raw assemblies are not yet public: K-PanRef, a Korean pangenome (28 haplotypes, 14 individuals) (Shin et al., 2026), and CPC, the Chinese Pangenome Consortium Phase 1 resource (116 haplotypes, 58 individuals) (Wang et al., 2026). Because neither project has released per-sample assembly FASTAs, their MHC sequences were instead pulled directly out of *their own* published Minigraph-Cactus graphs: for K-PanRef, every haplotype path was extracted from the GBZ with `vg paths -F`; for CPC, the CHM13 chr6:28–34 Mb reference nodes were extracted with `odgi`, and the span of every CPC haplotype walk through those nodes was read from the GFA. In both cases, extracted segments were kept only if they had a minimap2 asm20 alignment of at least 50 kb of matching bases and MAPQ ≥ 20 against the GRCh38 MHC. Three CPC samples that are also HPRC individuals were not

added twice. This gives 754 haplotype entries in total (Table 1); after removing the two single-haplotype references and five duplicate donor-assembly pairs (see below), 742 haplotypes from 371 name-reconciled donors remain for donor-level analyses. **Caveat:** K-PanRef and CPC haplotypes are graph paths, not original assemblies — Minigraph-Cactus clips sequence unaligned to the graph backbone, so haplotype-private insertions in these two cohorts can be shortened relative to the true assembly.

HPRC r2 was additionally split into four population strata using 1000 Genomes/HPRC release 2 sample metadata: HPRC-Japanese (JPT, 32 haplotypes), HPRC-Jewish (HG002 only, Ashkenazi, 2 haplotypes — a single individual, not a population estimate), HPRC-EastAsian (CHB/CHS/CDX/KHV and HG005, 70 haplotypes) and HPRC-Rest (360 haplotypes).

Table 1: Haplotypes analysed, by cohort/stratum.

Cohort / stratum	Haplotypes	Population	Source
APR	106	Arab (UAE)	Assembly
HPRC r2 – Japanese	32	Japanese (1000G JPT)	Assembly
HPRC r2 – Jewish	2	Ashkenazi (HG002 only)	Assembly
HPRC r2 – East Asian	70	CHB/CHS/CDX/KHV, HG005	Assembly
HPRC r2 – Rest	360	Mixed	Assembly
JaSaPaGe – Saudi	18	Arab (Saudi)	Assembly
JaSaPaGe – Japanese	20	Japanese (1000G JPT)	Assembly
K-PanRef	28	Korean	Graph path
CPC	116	Chinese	Graph path
GRCh38 / CHM13	2	Reference	Reference
Total	754		

Extraction and Immuannot annotation (unchanged core workflow)

The rest of the original CWL workflow (`hla_pangenome.cwl`) runs unchanged on the enlarged panel: per haplotype, `pgr-tk pgr-query` (Chin et al., 2023) extracts the extended MHC region (GRCh38 chr6:28,510,120–33,480,577 $\pm$ 100 kb), and Immuannot v3 (IPD-IMGT/HLA 3.55, IPD-KIR 2.13, RefSeq C4) (Zhou et al., 2024) calls gene structure and allele identity. All 754 haplotypes were re-aggregated, per-gene sequences re-cut, and `pgr-tk` bundle decompositions and `pgrb` (Garrison et al., 2024)/`odgi` (Guarracino et al., 2022) graphs rebuilt.

Whole-MHC Minigraph-Cactus pangenome graph

We built a single whole-MHC pangenome graph from all 754 haplotypes with `cactus-pangenome` (Cactus 3.3.0 static release; Minigraph-Cactus algorithm (Hickey et al., 2024)), referenced against both GRCh38 and CHM13, with `--gfa clip full --gbz clip full --giraffe clip --vcf --odgi full --viz`. The build ran on the NIG supercomputer (32 cores, 240 GB) in 3 h 43 min and produced a graph of 417,896 nodes and 579,137 edges (5.89 Mb of sequence), with `vg Giraffe` indexes for short-read alignment (Sirén et al., 2021).

Graph validation. We mapped one 1000 Genomes individual's (HG00096) MHC-region read pairs to the graph with `vg giraffe`: 654,059 of 657,052 pairs aligned (99.5%), 625,590 at `MAPQ` $\geq$ 20 (95.2%), in 26 s on 8 threads. This establishes that the graph is mappable to at high rates; it does not by itself establish that alignment to the graph improves genotyping, which we tested directly next.

Does graph-based genotyping beat a direct-read typer? A first, consensus-based attempt

As a first test of whether the new graph helps HLA typing at all, we built a pipeline that aligns 1000 Genomes short reads to the graph with `vg giraffe`, calls variants with `vg call`, phases them with read-backed WhatsHap (Patterson et al., 2015), takes a `bcftools consensus` hap1/hap2 FASTA per sample, and re-annotates that consensus with `Immuannot` — i.e., the graph is used to build a personal, phased pseudo-assembly, which is then typed exactly as the assembly panel above is typed. We compared this against T1K (Song et al., 2023), which types directly from the same raw reads without any graph or consensus step, and against real experimental truth (Gourraud et al. 2014's Sanger/SSO HLA typing of 1000 Genomes samples, 955 of which overlap this cohort (Gourraud et al., 2014)). We ran two independent T1K invocations on the same reads as a check on typer-level noise, and re-annotated the same consensus sequences against both the original frozen IPD-IMGT/HLA 3.55.0 and a freshly built 3.65.0 (the release T1K uses) to separate database-version effects from consensus-quality effects. Concordance was scored at 1-field (allele group) and 2-field (protein) resolution on unordered genotype pairs, on the 9 classical HLA genes (A/B/C/DRA/DRB1/DQA1/DQB1/DPA1/DPB1); `Immuannot`'s `:new` suffix (no exact database match) was stripped before truncation rather than treated as a literal field, and a genotype comparison was skipped rather than counted as a mismatch when fewer than 2 real fields remained on either side.

RCCX/C4 structural genotyping prototypes (exploratory, not the main typer)

In parallel, we prototyped two lightweight, held-out-validated approaches to genotype the RCCX module (which carries the C4A/C4B copy-number and long/short structural variation) directly from short reads without full HLA allele calling: a k-mer/marker-counting method matched against graph paths (`hla-structural/`, 106 donors: 18 pilot + 88 held out), and a dedicated 299-probe targeted dosage assay (`hla-targeted/`, benchmarked against the published tool C4Investigator (Marin et al., 2024) on 24 held-out donors, with C4Investigator's own output calibrated on 18 pilot donors before blind application to keep the comparison fair). Both used donor-level, family-disjoint held-out splits.

Classical 11-locus typing benchmark on the full panel

To check that the enlarged panel and pipeline still produce reliable classical HLA calls, we compared `Immuannot`'s exact-CDS two-field calls at 11 classical loci (A/B/C/DPA1/DPB1/DQA1/DQB1/DRB1/DRB3/DRB4/DRB5) on all eligible haplotypes of the 754-haplotype panel against the Gourraud et al. 2014 experimental Sanger/SSO truth at the 5 genes it covers (A/B/C/DRB1/DQB1), retaining published typing ambiguity and requiring two eligible current-CDS labels per donor.

The typer in progress: PanGenie genotyping directly against graph bubbles

The consensus-based pipeline above showed that going through an explicit per-sample consensus loses genotyping resolution (see Results). We are therefore building the actual typer around a different strategy, closer to how PanGenie (Ebler et al., 2022) and SpecHLA's personalized-reference approach avoid an explicit global consensus: genotype the graph's bubbles directly from k-mers in the raw short reads, per sample, without ever constructing a linear consensus sequence to re-annotate. Concretely: (1) `vg deconstruct` decomposes the whole-MHC graph into top-level bubbles relative to the GRCh38 path, giving a bubble-level VCF; (2) for 5 donor-disjoint, family-disjoint, population-balanced folds, we build two reference panels per fold — a “full” panel (all cohorts) and an “HPRC-only” panel — so that adding the Asian-specific haplotypes (APR, JaSaPaGe, K-PanRef, CPC) can be evaluated against an

HPRC-only baseline under the same held-out design used for the RCCX/C4 prototypes above; (3) PanGenie genotypes each held-out donor's short reads against both panels for its fold. The initial evaluation cohort is 40 1000 Genomes donors (20 East Asian, 20 South Asian), selected to be family-disjoint and population-balanced, with Arabian donors reserved for a later addition pending data availability. The same 40 donors' reads were also run through T1K and SpecHLA for comparison, following the design in our study plan ([hla-study-plan/plan.tex](#)), which further specifies HLA*LA and a long-read SpecImmune/HLAminer arm as later additions.

Status at the time of writing: panel construction (bubble VCF, 5-fold full/HPRC-only panels) and call generation are complete — all 120 jobs (40 donors $\times$ {T1K, SpecHLA, PanGenie-full, PanGenie-HPRC-only}) finished successfully with verified output integrity. Accuracy scoring against truth has not finished; we report the one completed sub-analysis (structural signature representation) below and describe the rest as ongoing work.

Results

The MHC still comes out complete after tripling the panel's non-HPRC diversity

747 of 752 non-reference haplotypes reach ≥ 0.99 MHC coverage, and 720 of 752 land on a single segment (Figure 1). The lower-coverage tail is concentrated in the two graph-derived cohorts, as expected from upstream clipping: 4 CPC haplotypes (lowest 0.92) and 1 JaSaPaGe-Saudi haplotype (0.96); all directly-assembled cohorts (APR, HPRC, JaSaPaGe-Japanese, K-PanRef) are at or near 1.00.

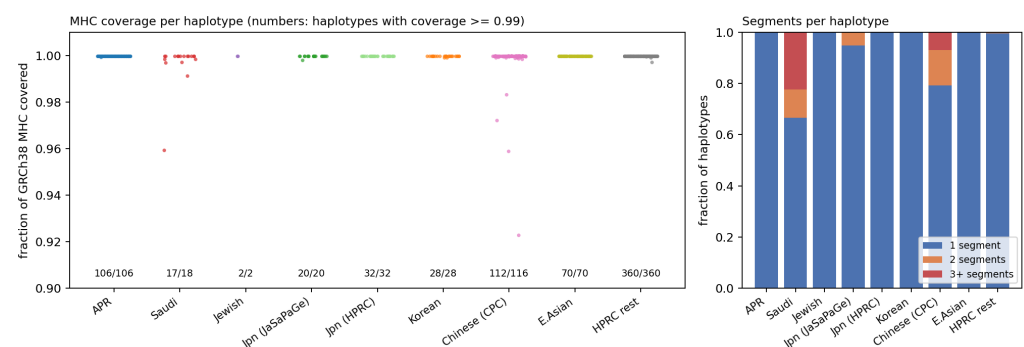


Figure 1: MHC region extraction across the 752 non-reference haplotypes of the enlarged, 9-stratum panel. Left: coverage of the GRCh38 MHC interval; numbers give haplotypes at coverage ($\geq$) 0.99. Right: number of extracted segments per haplotype. The graph-derived CPC and JaSaPaGe-Saudi cohorts show most of the incomplete/multi-segment cases.

Allele frequencies now separate nine strata, and the Korean/Chinese additions bring their own signal

Immuannot calls on the 9-stratum panel (Figure 2) place the two new cohorts sensibly alongside the others: HLA-A*11:01 reaches 29% in CPC-Chinese and 26% in other-East-Asian haplotypes, HLA-DRB1*12:02 16% in CPC-Chinese, alongside the previously seen A*24:02 (35–41% in Japanese strata) and DRB1*03:01 (18% in APR). DRB1 remains the gene with the largest gap between assembled sequence and the allele database: 57–79% of DRB1 gene copies have no full-length match in IPD-IMGT/HLA 3.55 across all nine strata.

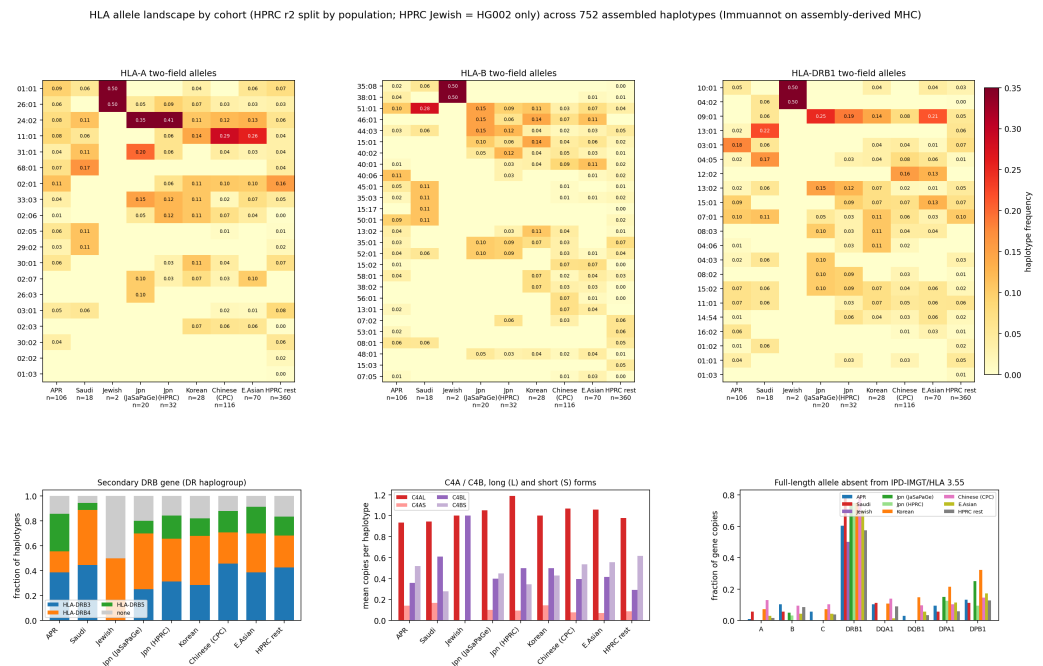


Figure 2: HLA allele landscape across the 752 assembled/graph-derived haplotypes, now split into 9 cohorts/strata (HPRC r2 divided by population). Top: HLA-A/-B/-DRB1 frequencies. Bottom: secondary DRB gene, C4A/C4B long/short form, and fraction of gene copies absent from IPD-IMGT/HLA 3.55. The HPRC-Jewish column is a single individual (HG002) and not a population estimate.

Two individuals are genuinely MHC-homozygous; one assembly holds the same haplotype twice

This finding is unchanged by the panel expansion, since it concerns specific HPRC/JaSaPaGe individuals (Figure 3): NA18976 and NA19909 carry fewer than 100 heterozygous 1000G SNPs per 100 kb across the 5 Mb MHC — assembly-independent evidence of genuine near-homozygosity — while JaSaPaGe's assembly of NA18952 shows full heterozygosity on *both* "haplotypes" and only 10 substitutions between them, confirming the same true haplotype was assembled twice (traced to a shared contig name with the public HPRC assembly of the same person).

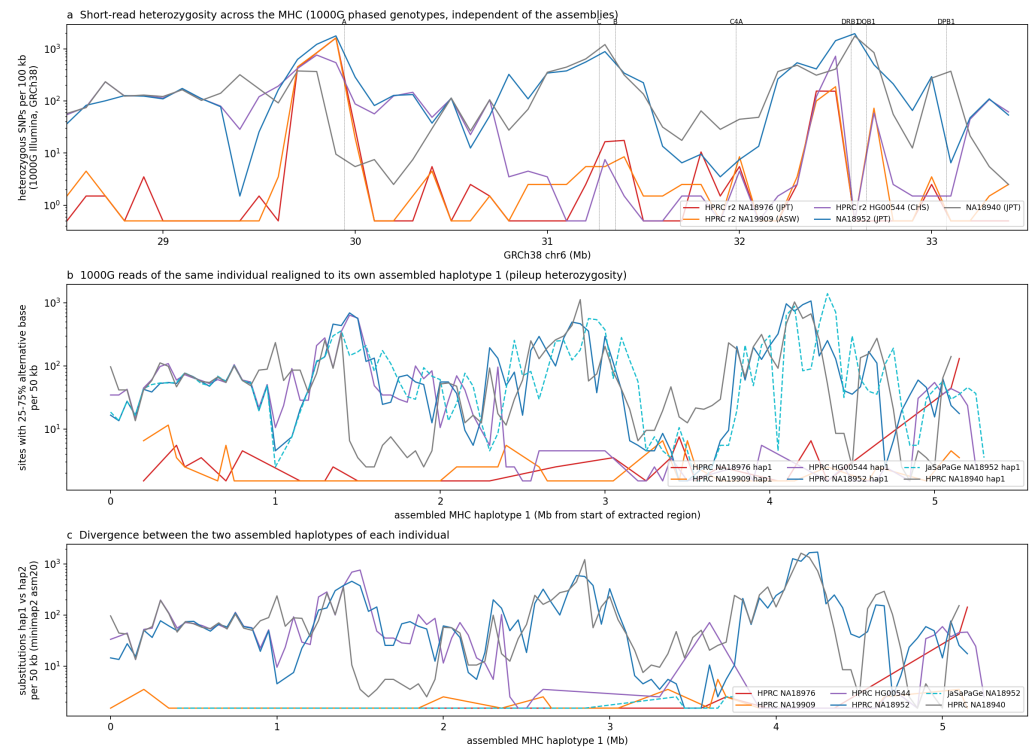


Figure 3: MHC homozygosity and haplotype-duplication check using 1000G data external to the assemblies.

A correction, and one genuinely novel coding allele left standing

Re-checking the ten classical-gene :new candidate alleles against the current IPD-IMGT/HLA 3.65 release resolved one of them: HG02717's HLA-DQB1 Ala→Asp change, previously reported as novel and independently read-supported, is in fact an exact match to DQB1*02:180:02 (registered in release 3.56) — the same allele the independent HPRC 4-field truth set of Lai et al. 2024 calls for this sample. NA20346's HLA-DPA1 Ala→Met change remains novel under 3.65, still backed by 24 of 51 of the individual's own 1000G reads (Figure 4).

a Candidate novel coding alleles (Immuannot 'new' with CDS differences) in classical HLA genes. private = variant 31-mer absent from the other 609 haplotypes of the 610-haplotype run; red = differences include indels (typical HiFi homopolymer assembly errors, treat as artefacts); green = heterozygous support in the individual's own short reads

haplotype	cohort	gene	closest known allele	coding difference(s) (known > assembly)	private variants	also on other hap	own 1000G Illumina reads (realigned to assembly)
apr003#1	APR	A	A*24:02:102:01	4 subst.: Val>Leu, Phe>Tyr, Ser>Pro, Lys>Asn	1	no	no public reads
apr011#1	APR	C	C*03:02:02:01	3 subst.: Ile>Leu, Ala>Gly, Gly>Cys	0	no	no public reads
apr001#1	APR	DRB1	DRB1*16:01:01:01	5 subst.: Gln>His, Val>Val, Leu>Arg, Met>Val, Arg>Arg	0	no	no public reads
apr038#1	APR	DRB1	DRB1*14:06:01	5 subst. + indels: Arg>Lys, Val>Ala, Arg>Arg, Cys>Cys, His>Gln	1	no	no public reads
apr002#1	APR	DQB1	DQB1*02:02:01:01	1 subst.: Leu>Leu	0	yes	no public reads
apr015#2	APR	DPB1	DPB1*10:01:01:01	1 subst.: Leu>Leu	1	no	no public reads
HG02717#1	HPRC-Rest (GWD)	DQB1	DQB1*02:02:01:01	Ala>Asp; = DQB1*02:180:02 (IMGT/HLA 3.65): not novel	0	no	8/14 assembly base, 6 other hap
NA20346#2	HPRC-Rest (ASW)	DPA1	DPA1*01:03:01:180	Ala>Met; novel in 3.65 (closest 03:02:02, 1 subst.)	1	no	24/51 assembly base, 27 other hap
ksa004#1	JaSaPaGe-Saudi	A	A*02:01:01:134Q	3 indels only (1-bp deletion, 2 insertions)	-	-	no public reads
HG03516#2	HPRC-Rest (ESN)	DPA1	DPA1*03:01:01:01	1-bp insertion only	-	-	-

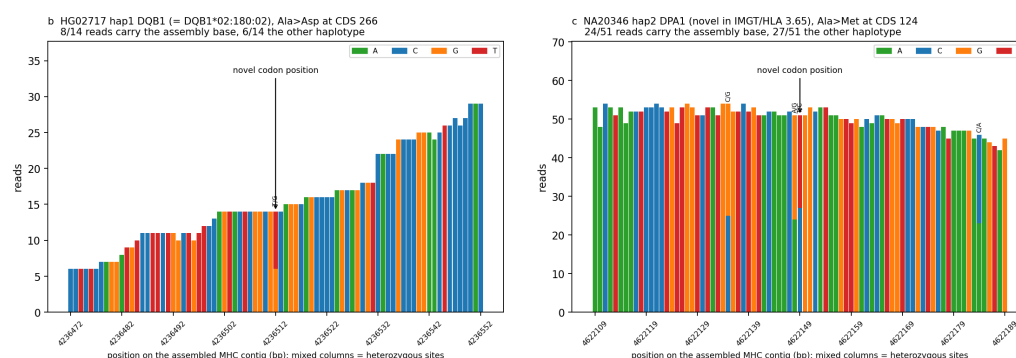


Figure 4: Candidate novel coding alleles and their support, rechecked against IPD-IMGT/HLA 3.65.

The graph-consensus pipeline is as accurate as a direct-read typer, but resolves far fewer genotypes

This is the key methodological result for the typer we are building. Immuannot's old and new database re-annotations of the same graph-derived consensus agree closely with each other (91% at 1-field, 89% at 2-field, pooled over the 9 classical genes; Figure 5) — expected, since only the reference database changed. T1K agrees with either Immuannot re-annotation only about half the time at 2-field (49–50%), confirmed to be a real, reproducible gap by a second, fully independent T1K run rather than an artefact of one invocation (80% concordance between the two T1K runs on the same 9 genes). HLA-DRB1 concordance collapses to 10–25% in every pairwise comparison, including old-DB-vs-new-DB on the identical consensus sequence — implicating consensus/phasing quality at this paralog-rich locus, not database version or typer choice.

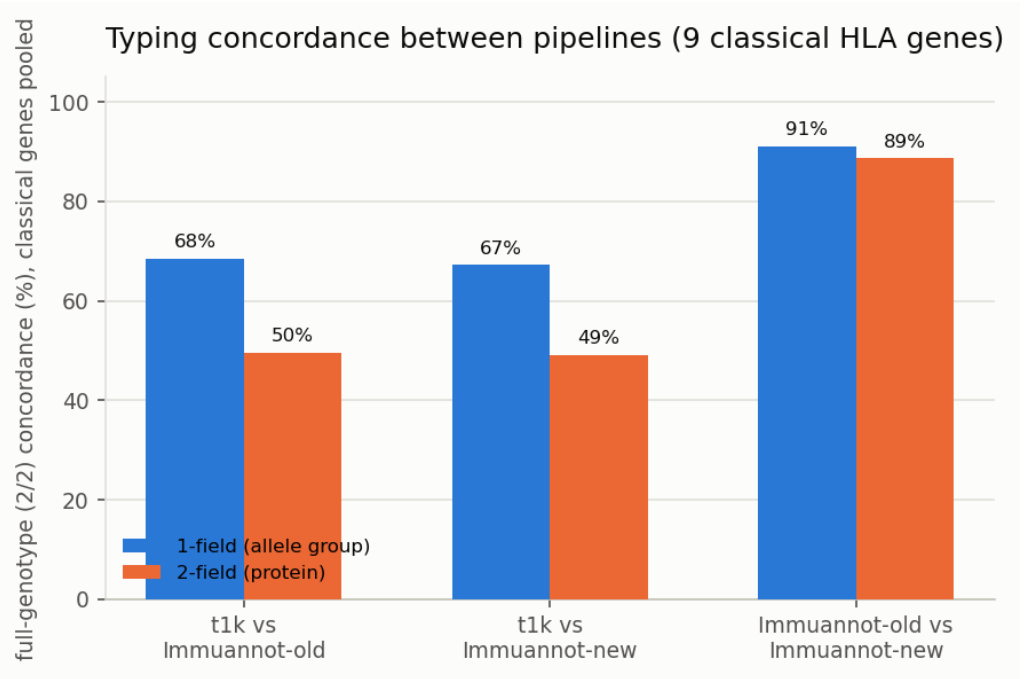


Figure 5: Pairwise typing concordance between the graph-consensus pipeline (Immuannot, two database versions) and a direct-read typer (T1K, two independent runs), pooled over 9 classical HLA genes.

Checked against real experimental truth (Figure 6), T1K and the graph-consensus pipeline are statistically tied on accuracy when the latter actually commits to an answer: 98.8% (T1K, $n = 1,843$ checkable genotypes) vs. 98.5% (Immuannot on graph consensus, either database version, $n = 613$). The gap is in *resolution*, not correctness: the graph-consensus pipeline only reaches a checkable 2-field call for 33% of eligible genotypes, against T1K's 90%. **This is why we are not simply scaling up the consensus-based approach:** going through a single per-sample consensus sequence costs coverage on exactly the genes (DRB1 above all) that most need a population-specific reference, which is the opposite of what this project is for. It directly motivates genotyping graph bubbles per-sample with PanGenie instead of collapsing to one consensus (Methods, "The typer in progress").

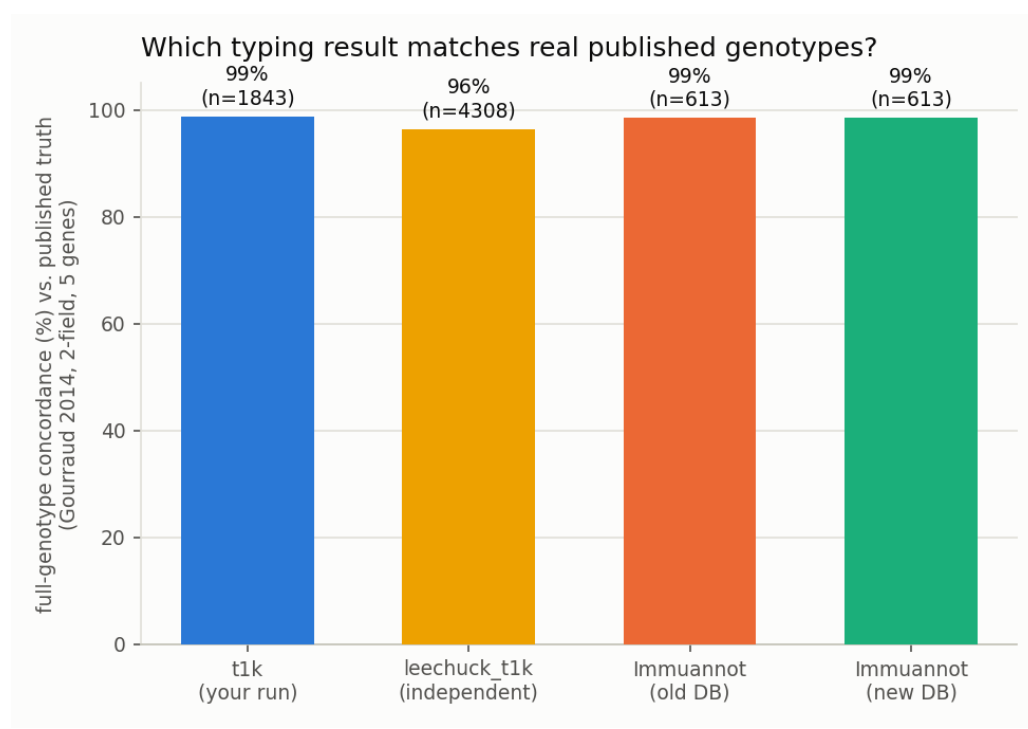


Figure 6: Genotype concordance against real published truth (Gourraud et al. 2014, 2-field, 5 classical genes): T1K and the graph-consensus pipeline are tied on accuracy but resolve very different fractions of eligible calls.

RCCX/C4 structural prototypes did not beat trivial baselines

Both held-out-validated prototypes for RCCX/C4 structural genotyping were informative but did not demonstrate a benefit from graph structure over simpler baselines. The marker-counting prototype recovered RCCX gene content trivially (106/106, as expected — DRB copy number is easy) but only 38–64 of 106 full structural-signature pairs correctly, and calibrated plain reference read depth tied its best copy-number variant, so no graph-specific benefit was established. The targeted 299-probe assay matched but did not beat a plain-depth baseline on total C4 dosage (24/24 for both) and scored 21/24 on full RCCX structural pairs — slightly worse than a pre-existing simpler method's 22/24. We report this as a genuine, disclosed negative result rather than omit it: it shows that RCCX/C4 structure is genuinely hard to call from short reads regardless of method, which is useful context for anyone else building an HLA/MHC structural typer on this or similar panels.

Classical calls on the enlarged panel remain concordant with real truth

Against Gourraud et al. 2014, exact-CDS two-field calls on the full 754-haplotype panel agree with experimental truth at 62/64 (HLA-A), 64/64 (HLA-B), 58/63 (HLA-C), 59/63 (HLA-DRB1) and 58/61 (HLA-DQB1). Of the 14 discrepancies, 10 are compatible with the historical assay's antigen-binding-exon resolution under frozen G-group definitions, and of the 4 remaining, 3 have independent read or cross-assembly support for the assembly's own call (NA18943 A/DRB1, NA19007 A); one (NA18608 DRB1) is unresolved because the relevant probes are non-unique. This benchmark, together with the cross-method comparison in Figure 7 (FuFiHLA vs. Immuannot on the same assemblies, 92–100% 2-field agreement across 373–376 individuals; T1K vs. either method on the 23 assembled individuals typed by T1K so far), supports that the extraction+annotation pipeline itself remains reliable at the enlarged scale — the resolution problem identified above is specific to the graph-consensus arm, not to Immuannot or the panel in general.

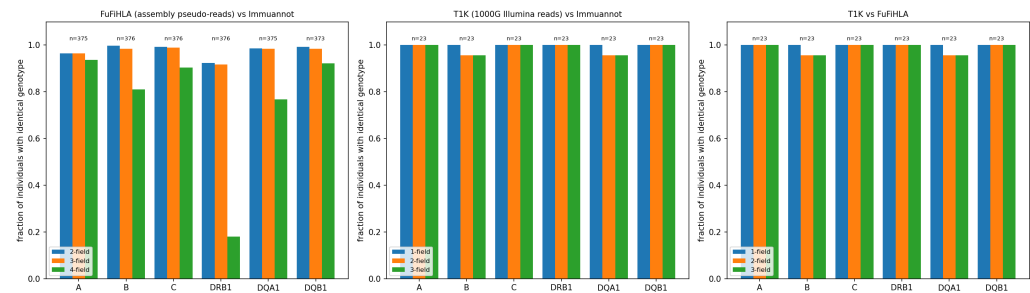


Figure 7: Typing concordance between FuHiHLA, Immuannot and T1K on the same assembled individuals, at 1–4 field resolution.

The PanGenie-based typer: infrastructure complete, accuracy scoring in progress

Bubble-level panel construction (5 folds $\times$ {full, HPRC-only} arms) and call generation for the initial 40-donor East + South Asian cohort are complete: all 120 jobs (T1K, SpecHLA, PanGenie-full, PanGenie-HPRC-only, per donor) finished successfully with verified output integrity (matching file counts and record counts for all outputs, including 3,250,704 VCF records from the full-panel PanGenie arm vs. 3,504,632 from the HPRC-only arm). Accuracy scoring against truth has not been run yet and is explicitly future work (see Discussion). One sub-analysis is complete and worth reporting now, honestly, as a null result: for the RCCX and DRB structural signature classes present among these 40 donors, the full panel and the HPRC-only panel represent exactly the same classes (100/100 for DRB, 95/100 for RCCX) — the 144 Asian-specific graph-derived haplotypes added no additional *represented* structural diversity for this particular set of held-out donors. This does not mean the Asian-specific haplotypes are uninformative in general (the population allele-frequency results above show clear population-specific signal); it means that, at least for this initial cohort and this particular representation metric, we have not yet demonstrated the benefit we are looking for, and accuracy scoring — not representation counting — is the test that will actually tell us if the Asian-specific panel helps.

Discussion

Three things are real and usable now: an Asian- and Arab-enriched HLA haplotype panel that has essentially tripled in non-HPRC diversity since our first week (754 haplotypes, 9 population strata, extending to Korean and Chinese cohorts via their own pang genome graphs where raw assemblies are not yet public); a whole-MHC Minigraph-Cactus graph built from that panel with validated short-read mapping; and a rigorous demonstration of *why* the obvious first way to use that graph for typing — align, call, phase, consensus, annotate — does not beat an existing direct-read typer (T1K) on accuracy, and specifically loses resolution at exactly the paralog-rich loci (DRB1 above all) that most need a better reference. That negative result is not a detour from the goal of “building a better HLA typer using pang genomes” — it is the finding that redirected the typer’s design away from an explicit consensus step and toward genotyping graph bubbles directly with PanGenie, closer to how SpecHLA avoids a single global reference. That typer is still in progress: the bubble-panel infrastructure and initial 40-donor (East + South Asian) call generation are done and verified, but accuracy scoring against real truth — the actual test of whether the Asian-specific panel produces better HLA calls than an HPRC-only one — has not been completed and is the immediate next step. We plan to report those numbers as a follow-up once scoring finishes; readers should treat “the typer” as a documented work-in-progress here, not a finished, benchmarked method. The two RCCX/C4 structural genotyping prototypes we built alongside this work, and reported honestly as not

beating trivial baselines, are a related but separate exploration, useful mainly as evidence that RCCX/C4 structure is hard to call from short reads regardless of approach.

Immediate next steps, in order: (1) finish accuracy scoring of the 40-donor PanGenie/T1K/SpecHLA comparison, full panel vs. HPRC-only, against Gourraud 2014 and any other available truth for these donors; (2) extend the benchmark cohort to the Arabian stratum once data access allows, per the original study design; (3) add HLA*LA and the long-read SpecImmune/HLAminer arm specified in the study plan; (4) investigate whether K-PanRef/CPC's graph-derived (rather than assembled) haplotypes constrain the bubble panel less than we might expect, given the upstream-clipping caveat noted in Methods; (5) revisit HLA-DRB1 phasing specifically, since it is the worst-performing locus in every comparison we ran, including database-version-only comparisons on an identical consensus sequence.

Acknowledgements

We thank the organisers of DBCLS BioHackathon Japan 2026 and the NIG supercomputer BioHackathon node for compute access; the Arab Pangenome Reference (MBRU) (Nassiri et al., 2025), K-PanRef (Shin et al., 2026), CPC (Wang et al., 2026), HPRC and JaSaPaGe project teams for making assemblies and pangenome graphs available; and the authors of pgr-tk (Chin et al., 2023), Immuannot (Zhou et al., 2024), pggg (Garrison et al., 2024), odgi (Guarracino et al., 2022), minimap2 (Li, 2018), Cactus/Minigraph-Cactus (Hickey et al., 2024), vg/Giraffe (Sirén et al., 2021), WhatsHap (Patterson et al., 2015), PanGenie (Ebler et al., 2022), T1K (Song et al., 2023) and C4Investigator (Marin et al., 2024), on which this work is built.

References

Author contributions

First Author: Writing – original draft; Last Author: Conceptualization, Writing – review & editing.

References

- Chin, C.-S., Behera, S., Khakhal, A., Sudmant, P. H., Wagner, J., Chin, J., Peluso, P., Timp, W., Zook, J. M., & Schatz, M. C. (2023). Multiscale analysis of pangenomes enables improved representation of genomic diversity for repetitive and clinically relevant genes. *Nature Methods*, 20, 1213–1221. <https://doi.org/10.1038/s41592-023-01914-y>
- DeepOmics Lab. (2023). Spechla: Full-resolution hla typing from ngs data with a personalized reference graph.
- Dilthey, A. T., Mentzer, A. J., Carapito, R., Cutland, C., Cereb, N., Madhi, S. A., Rhie, A., Koren, S., Bahram, S., McVean, G., & Phillippy, A. M. (2019). Hla*la – hla typing from linearly projected graph alignments. *Bioinformatics*. <https://doi.org/10.1093/bioinformatics/btz235>
- Ebler, J., Ebert, P., Clarke, W. E., Rausch, T., Audano, P. A., Houwaart, T., Mao, Y., Korbel, J. O., Eichler, E. E., Zody, M. C., Dilthey, A. T., & Marschall, T. (2022). Pangenome-based genome inference allows efficient and accurate genotyping across a wide spectrum of variant classes. *Nature Genetics*, 54, 518–525. <https://doi.org/10.1038/s41588-022-01043-w>
- Garrison, E., Guarracino, A., Heumos, S., Villani, F., Bao, Z., Tattini, L., Hagmann, J., Vorbrugg, S., Marco-Sola, S., Kubica, C., Ashbrook, D. G., Thorell, K., Rusholme-Pilcher, R. L., Colonna, V., Formenti, G., Mwaniki, S., Erb, I., Lehmann, N., Prins, P., ... Paten, B. (2024). Building pangenome graphs. *Nature Methods*. <https://doi.org/10.1038/s41592-024-02430-3>
- Gourraud, P.-A., Khankhanian, P., Cereb, N., Yang, S. Y., Feolo, M., Maier, M., Rioux, J. D., Hauser, S., & Oksenberg, J. (2014). Hla diversity in the 1000 genomes dataset. *PLOS ONE*, 9(7), e97282. <https://doi.org/10.1371/journal.pone.0097282>
- Guarracino, A., Heumos, S., Nahnsen, S., Prins, P., & Garrison, E. (2022). Odgi: Understanding pangenome graphs. *Bioinformatics*, 38(13), 3319–3326. <https://doi.org/10.1093/bioinformatics/btac308>

- Hickey, G., Monlong, J., Ebler, J., Novak, A. M., Heller, D., Genthaler, L., Marschall, T., & Paten, B. (2024). Pangenome graph construction from genome alignments with minigraph-cactus. *Nature Biotechnology*, 42, 663–673. <https://doi.org/10.1038/s41587-023-01793-w>
- Ito-Naito, T., et al. (2026). Genetic diversity and haplotype architecture of non-classical hla class i genes in the japanese population. *HLA*, 108(2), e70898.
- Li, H. (2018). Minimap2: Pairwise alignment for nucleotide sequences. *Bioinformatics*, 34(18), 3094–3100. <https://doi.org/10.1093/bioinformatics/bty191>
- Marin, R. S., Augusto, D. G., Wade, K., & Hollenbach, J. A. (2024). High-throughput complement component 4 genomic sequence analysis with c4investigator. *HLA*, 103. <https://doi.org/10.1111/tan.15273>
- Nassir, N., Mohamed Ibrahim, S., Alsafar, H., et al. (2025). A draft uae-based arab pangenome reference. *Nature Communications*. <https://doi.org/10.1038/s41467-025-61645-w>
- Patterson, M., Marschall, T., Pisanti, N., van Iersel, L., Stougie, L., Klau, G. W., & Schönhuth, A. (2015). Whatshap: Weighted haplotype assembly for future-generation sequencing reads. *Journal of Computational Biology*, 22(6), 498–509. <https://doi.org/10.1089/cmb.2014.0157>
- Shin, S., et al. (2026). A korean pangenome reference of 14 healthy individuals supports structural variant analysis in disease genomes. *medRxiv*. <https://doi.org/10.64898/2026.07.06.26357367>
- Sirén, J., Monlong, J., Chang, X., Novak, A. M., Eizenga, J. M., Markello, C., Sibbesen, J. A., Hickey, G., Chang, P.-C., Carroll, A., Gupta, N., Gabriel, S., Blackwell, T. W., Ratan, A., Taylor, K. D., Rich, S. S., Rotter, J. I., Haussler, D., Garrison, E., & Paten, B. (2021). Pangenomics enables genotyping of known structural variants in 5202 diverse genomes. *Science*, 374(6574), abg8871. <https://doi.org/10.1126/science.abg8871>
- Song, L., Bai, Y., Nyunt, Y. M., Zhang, Y., Zhang, N., & Hu, X. (2023). Efficient and accurate genotyping of kir and hla genes from massively parallel sequencing data. *Genome Research*. <https://doi.org/10.1101/gr.277585.122>
- Wang, T., et al. (2026). A pangenome reference of 1000 chinese genomes. *Nature*. <https://doi.org/10.1038/s41586-026-10315-y>
- Zhou, Y., Rich, S. S., Onengut-Gumuscu, S., & Chin, C.-S. (2024). Immuannot: An efficient and accurate annotation tool for genes in the human immune region. *Genome Research*. <https://doi.org/10.1101/gr.278985.124>

Appendices

Workflow, analysis scripts, full figure captions and result tables referenced in this report are in this project's companion repository, `pangenome-bh26` (github.com/leechuck/pangenome-bh26): the extraction/annotation workflow and its README (`hla/`), the whole-MHC graph build and its README (`hla/docs/MHC_GRAPH_README.md`, `hla/nig/mhc_mc_graph.sbatch`), the graph-consensus-vs-T1K evaluation (`hla/analysis/gc_*.py`, `hla/results/slides/gc_typing_concordance.tex`), the RCCX/C4 structural prototypes (`hla-structural/`, `hla-targeted/`), the classical 11-locus benchmark (`hla-analysis/FINAL_REPORT.md`), and the PanGenie-based typer under development (`hla-study-plan/plan.tex`, `hla-asian50/`).